Check for updates

Received: 21 October 2021 | Accepted: 17 March 2022

DOI: 10.1111/1365-2745.13898

Journal of Ecology | BRITISH ECOLOGICAL SOCIETY

RESEARCH ARTICLE

Towards a better ecological understanding of metacommunity stability: A multiscale framework to disentangle population variability and synchrony effects

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Funding information
Grantová Agentura České Republiky, Grant/Award Number: 20-02901S

Handling Editor: David Gibson

Abstract

1. Despite great progress in our understanding of the mechanisms governing ecosystem stability in local communities, we still lack knowledge at a larger spatial scale. Studying the stability of metacommunities requires assessing the temporal stability and synchrony of populations across space and organizational levels. Previous attempts to disentangle these effects have provided limited ecological interpretations, and conceptual improvements are needed to identify the underlying ecological processes.
2. We propose an extended framework aiming at disentangling simultaneously the relative effects of population stability and different types of synchronies on metacommunity stability. We adapted previous methods of decomposing stability into a new set of indices associated with clearer ecological hypotheses. Particularly, we provide synchrony indices that are not affected by statistical properties of the metacommunity but focus on species responses to environment, demography and interactions. We applied this framework to a unique dataset describing the sorted biomass of individual plant populations, across 12 communities of a species-rich meadow, and for 16 years. The communities were sampled in different treatments of fertilization and dominant removal to evaluate the effect of environmental heterogeneity on stability.
3. We found higher stability at a larger spatial scale, mainly due to statistical averaging (portfolio effect). The variability of individual populations was an important determinant of the stability of the whole metacommunity. Consistent with the hypothesis of a common response to environmental conditions, we found that the fluctuations of populations were mostly synchronized (within and between species) at a large spatial scale and tended to destabilize the metacommunity. On the other hand, opposite fluctuations (anti-synchrony) between populations occurred at the local scale, probably due to local species interactions.
4. *Synthesis*. Our framework appears as a powerful tool to test how ecological processes occurring simultaneously at different spatial and organizational scales affect the stability of metacommunities. This study advances our

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ecological understanding of the processes underlying the stability of species-rich environments.

KEYWORDS
compensatory dynamics, metacommunity, metapopulation, plant population and community dynamics, population synchrony, population variability, spatial stabilization, temporal stability

1 | INTRODUCTION

Ecosystem stability through time, or constancy (sometimes called invariability, see Box 1 in de Bello et al., 2021 for definitions), depicts an ecosystem's ability to buffer the effects of the natural fluctuations in environmental conditions, and maintain consistent functioning (Pimm, 1984). What factors drive such temporal stability is a long-standing key question in ecology (Lepš et al., 1982; Loreau & de Mazancourt, 2013; MacArthur, 1955; May, 1974; Tilman, 1996; Turner et al., 1993), which yet remains poorly elucidated, especially at a large spatial scale (Wang & Loreau, 2014). At the community level, there is an extensive body of theoretical frameworks (e.g. de Mazancourt et al., 2013; Ives et al., 1999; Loreau & de Mazancourt, 2008) and empirical results (Doležal et al., 2020; Fischer et al., 2020; Lepš et al., 1982; Tilman et al., 2006) examining the effect of diversity, and biotic or abiotic forcing factors (e.g. competition, resource availability, disturbance level) on the temporal stability of community productivity. Indeed, it is assumed that these factors could influence one or both underlying components of the community stability, namely, (1) the stability of individual species composing the community and (2) the compensatory dynamics between the fluctuations of these species. The latter, which results from asynchrony between species (degree of temporal desynchronization) partly compensating for their fluctuations, has often been considered as a major stabilizing process (Gonzalez & Loreau, 2009; Valencia et al., 2020).

With the increasing use of the metacommunity concept (Leibold et al., 2004; i.e. a set of local communities that are linked by dispersal of multiple interacting species: Wilson, 1992), other factors such as spatial diversity (β and γ diversity) and environmental heterogeneity are now commonly included in the analyses of stability (Wang et al., 2019; Wang & Loreau, 2014; Wilcox et al., 2017). In these studies, the stability of the metacommunity is decomposed into indices of stability and synchrony calculated either at different spatial (within and between communities) or organizational levels (within and between species; Wilcox et al., 2017). Recently, Hammond et al. (2020) highlighted the weaknesses of such an approach and proposed a unified framework to study the stability of metacommunities. This framework (Hammond et al., 2020) differs from previous work (e.g. Wang et al., 2019; Wilcox et al., 2017) by considering the metacommunity as a set of local populations instead of local communities or metapopulations separately, allowing them to compare the relative importance of compensatory dynamics at different spatial and organizational scales simultaneously. It is based on the mathematical partitioning of the metacommunity variability (reciprocal of stability estimated by the squared coefficient of variation) into three additive indices representing the compensations of the fluctuations of (1) δ: different species within communities, (2) β_{MP}: the same species across communities and (3) β_{CC}: different species across communities (see Box 1 for details). This approach revealed that the compensation between species across communities (β_{CC}) has probably been underestimated so far (termed 'hidden source of stability' in Hammond et al., 2020) proving the relevance of testing all types of stabilization simultaneously. Their analysis shows that β_{CC} was the dominant stabilizing effect in a tropical rock pool metacommunity. They primarily interpreted this result as an effect of environmental heterogeneity across plots, disrupting the synchrony between distant populations, and thus leading to spatial compensation at the metacommunity level (Hammond et al., 2020).

Here we propose a framework adapted from Hammond et al. (2020) with a similar partitioning of the metacommunity squared coefficient of variation. In our opinion, the framework from Hammond et al. (2020) provides limited ecological interpretations and can still be improved by two major modifications:

First, it is known that the level of compensation between populations (sensu Gonzalez & Loreau, 2009) is due to the degree of synchrony between each population pairs—which relates to various ecological processes (see Table 1 for details)—but also to the statistical averaging (see Doak et al., 1998)—which primarily depends on the total number of populations (Thibaut & Connolly, 2013; Appendix S1). Although being good descriptors of the overall compensatory dynamics, the indices presented in Hammond et al. (2020) do not differentiate these components (similarly to all indices related to the φ index introduced by Loreau & de Mazancourt, 2008, see Box 1). The statistical averaging has little influence on the results when comparing systems with a similar number of populations. However, this distinction is particularly needed when studying metacommunities, since there are often many more pairs of populations across communities than within communities. For example, Hammond et al. (2020) acknowledge that they cannot distinguish whether the high compensation between species across communities (β_{CC}) found in their study is associated with an ecological process affecting synchrony patterns between populations, or due to the high number of populations considered in this term (Box 1, Appendix S1). Furthermore, ecological hypotheses related to species synchrony commonly predict deviation from independent fluctuations (Table 1). For example, a common response to environmental conditions should result in coordinated fluctuations (positive synchrony), while competitive interactions should be associated with opposite fluctuations (negative synchrony) (Table 1). Hence,

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TABLE 1 Expected synchrony between plant population productivity at different organizational (within and between species) and spatial (within and between communities) scales in the context of a metacommunity. The ecological processes expected to cause an overall positive synchrony (sync), independent fluctuation (Ind) or anti-synchrony (anti-sync) are given. Environmental heterogeneity between communities represents variations in soil characteristics and resource availability. Here we assume that the communities experience similar weather fluctuations and same disturbance regime. Otherwise, the type of synchrony may be hard to predict. More details or examples in each case are given in the table footnote. Coloured circles refer to the related synchrony indices in our framework (Equation 4 and Figure 1): Orange $Pop.sync_{intra}$; yellow $Pop.sync_{direct}$; blue $Pop.sync_{indirect;}$ dark blue $Pop.sync_{no}$

	Within communities	Between communities in homogeneous environment	Between communities in heterogeneous environment
Within species		**Sync**: Common response of populations to environmental fluctuations[a] **Ind**: Demographic stochasticity[b]	**Ind or Anti-sync**: Independent or opposite response to environmental fluctuations[c]
Between species	**Sync**: Common response of species to environmental fluctuations[d] **Ind**: Demographic stochasticity[b] **Ind**: Different response of species to environmental fluctuations[e] **Ind or Anti-sync**: Different effect of higher trophic levels, particularly species-specific pathogens or herbivores[f]		
	Anti-sync: Direct competition[g]	**Anti-sync**: Indirect competition[h]	
			Ind or Anti-sync: Independent or opposite response to environmental fluctuations[c]

[a]Trait similarity within species should lead to similar responses to weather fluctuations.

[b]The demographic stochasticity effect is strong for species with a low number of independent modules (e.g. ramets, individuals) and a strong effect of the life stage on the annual productivity. Its effects decrease with the increasing number of independent modules considered (due to the law of large numbers). It also decreases with the spatial scale because the number of individuals increases with area.

[c]Depending on local conditions (e.g. soil type, elevation), weather fluctuations can be favourable or unfavourable for plant growth. For example, a warm year can be beneficial for growth or create drought conditions that repress growth depending on the soil field capacity for water and access to groundwater.

[d]Favourable conditions (e.g. warm, and wet years) should enhance growth while unfavourable conditions (e.g. cold, and dry years) should repress growth for all species.

[e]Trait dissimilarity between species results in (1) different responses to environmental fluctuations due to differences in resource access (e.g. deep rooting allows drought resistance during dry years) or (2) same responses to environmental fluctuations but with different timing (after a disturbance, fast-growing species can recover faster than slow-growing species, resulting in apparent independent productivity).

[f]Wet years can be favourable for growth, but also fungal pathogens. This might lead to high growth of resistant species but low growth of susceptible species.

[g]Physiological competition for resources or release of the competitive pressure during years with unfavourable conditions can create anti-synchronous dynamics.

[h]Under low species turnover between communities, populations of distinct species in distant communities can have anti-synchronous fluctuations if the two species are locally competing and have intraspecific synchrony between communities.

it is of interest to use synchrony indices that distinguish signals of positive (partial synchrony), null (independent fluctuation) or negative (anti-synchrony) synchrony. Overall, we propose a framework that focuses on the compensations between populations that relate to ecological processes and not to the statistical properties of the metacommunity. To do so, we use independent fluctuations rather than perfect synchrony as a benchmark for the variation partitioning (similar to, e.g. Gross et al., 2014; Schluter, 1984; Vasseur et al., 2014). Our partitioning allows calculating (1) the expected metacommunity variability under the hypothesis of independent fluctuations between populations and (2) indices describing positive, independent, or negative synchrony between populations at different spatial and organizational scales. All terms of our partitioning approach include compensatory dynamics: the first term evaluates the effect of individual population variability, but also the overall averaging effect due to the number of populations in the metacommunity. The other terms focus on the compensations between populations due to synchrony patterns and can be associated with explicit ecological hypotheses. Overall, we believe that this benchmark modification provides a better understanding of the ecological processes driving stability.

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BOX 1 Definitions and estimation methods of the synchrony effect on the metacommunity stability

Hammond et al. (2020) proposed a unified framework to estimate the effect of compensatory dynamics on metapopulation stability. This framework describes a metapopulation as a set of *n* local populations (independently of the species identity) instead of considering sets of local communities or metapopulations separately, like in previous works (Wang et al., 2019; Wilcox et al., 2017). It allows comparing simultaneously the relative importance of fluctuation compensations at different spatial (within and between communities) and organizational scales (within and between species). Three indices of population asynchrony describing compensatory dynamics (δ: between species within communities, β_{MP}: within metapopulations, and β_{CC}: between species and between communities) are derived from the following equation:

$$CV_M^2 = \left(\sum_{i=1}^{n} p_i CV_i \right)^2 - (\delta + \beta_{MP} + \beta_{CC})$$

where CV_M^2 is the metacommunity squared coefficient of variation, p_i and CV_i are, respectively, the relative abundance to the total metacommunity and the coefficient of variation of population *i*. The first term of the equation (the squared sum) corresponds to the expected CV_M^2 under perfect synchrony of all populations and the three terms of asynchrony (δ, β_{MP} and β_{CC}) quantify the deviation from this value, corresponding to stabilizing processes due to compensations at different spatial and organizational scales. δ, β_{MP} and β_{CC} are null when populations are perfectly synchronized and strictly positive otherwise. Their values depend on synchrony patterns but also on the number of populations considered and the individual population variability (see Appendix S1). When comparing metacommunities with a different number of communities, species richness or with populations of different variability, these indices poorly describe the synchrony between population temporal fluctuations.

Table 1 summarizes common ecological hypotheses related to population synchrony at different spatial and organizational scales. They should be tested using indices not affected by the metacommunity structure (Appendix S1). We therefore propose four new indices that focus on population synchrony ($Pop.sync_{direct}$: between species within communities, $Pop.sync_{intra}$: within metapopulations, and $Pop.sync_{indirect}$ and $Pop.sync_{no}$: between species and between communities with or without indirect interaction—see the material and method section for more details) based on the following equation:

$$CV_M^2 = \sum_{i=1}^{n} (p_i CV_i)^2 + \text{Pop. sync}_{direct} + \text{Pop. sync}_{intra} + \text{Pop. sync}_{inderect} + \text{Pop. sync}_{no}$$

Here, the first term of the equation (the sum of squares) corresponds to the expected CV_M^2 under independent fluctuations of all populations and the *Pop.sync* components quantify the overall synchrony between the corresponding populations. These values can be negative, null or positive depending on the prevalent type of synchrony (anti-synchrony, independent fluctuation or partial synchrony respectively) in the corresponding group of populations. Hence, we argue that our framework relies on closer links between mathematical indices and specific ecological mechanisms. In our case, positive synchrony is associated with destabilization and negative synchrony is considered as stabilizing compared to independent fluctuations. Note that δ and $Pop.sync_{direct}$, β_{MP} and $Pop.sync_{intra}$, β_{CC} and $Pop.sync_{indirect}$ and $Pop.sync_{no}$ are considering the same pairs of populations and can be compared directly.

Second, we propose identifying two categories of cross-community synchrony between species to improve our ecological understanding of the observed patterns. Indeed, one can expect these species synchrony to be mitigated by the species turnover between communities. Under high species turnover, we can assume that populations of different species across communities do not interact and respond independently from each other's fluctuations. However, under low species turnover, species in different plots may also coexist within plots and, if their metapopulations are synchronized, they can have indirect effects on each other's fluctuations. Instead of considering populations with or without indirect interactions in a unique index (β_{CC}), we propose to evaluate their synchrony patterns separately. The calculated values can be related to different ecological processes: the synchrony of populations under indirect interactions assesses the intensity of indirect competition across communities, especially compared to the synchrony values within communities (Table 1), while the synchrony of non-interacting populations describes the overall coordination of species responses to environmental conditions (Table 1).

Our framework aims at decomposing the overall metacommunity variability into additive terms representing the relative effects of (1) the variability of individual populations and (2) four categories of populations synchrony (among species within communities, within species between communities, among species between communities under indirect interactions and without indirect interaction). Although each

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type of population synchrony can be associated with a corresponding asynchrony index in Hammond et al. (2020) (Box 1), only our framework allows testing the ecological hypotheses in Table 1 (Appendix S1). Specifically, we test whether populations deviate from independent fluctuations, towards positive synchrony within metapopulations due to their common responses to the environment, and negative synchrony between interacting populations due to competition (Table 1). Also, we aim at estimating the contribution of the individual population variability and each synchrony type to the overall stability. We address these questions across different spatial scales using data from a long-term experiment monitoring the productivity of a central European meadow (Lepš, 2004, 2014, 2018). These data consist of the above-ground biomass of individual species populations, at the peak of the vegetation season, for 16 years and across 12 plots representing four treatments of fertilization and dominant removal. The experiment is considered as a metacommunity including communities in a homogeneous (within treatments) and a heterogeneous space (between treatments). The design provides a unique opportunity to explicitly integrate environmental heterogeneity in the partitioning of the metacommunity variability, further exploring the drivers of the 'hidden source of stability' between communities described by Hammond et al. (2020). More precisely, we applied our framework to address two major questions:

1. Using different subsets of plots, what is the effect of the spatial scale (local community, treatment, metacommunity) on the stability of productivity, and what is the relative effect of individual population variability and population synchrony at each scale? Do we observe the same patterns at the intraspecific level using a selection of four abundant species?
2. Using all plots, which type of population variability and synchrony matters for the metacommunity stability?

We expect higher stability at a larger spatial scale. Indeed, based on the "portfolio effect" proposed by Doak et al. (1998), we expect a higher averaging of population fluctuations when increasing the number of populations considered. However, the variation of the overall synchrony with space is hard to predict since positive, independent or negative synchrony can be expected at each scale (Table 1). At the metacommunity level, we expect positive synchrony within metapopulations, prevailing negative synchrony between interacting populations, and independent or positive synchrony between non-interacting populations (see Table 1 for detailed hypotheses). We expect that these general patterns will be affected by the treatments since fertilization and competitor removal can modify the intensity of species interactions.

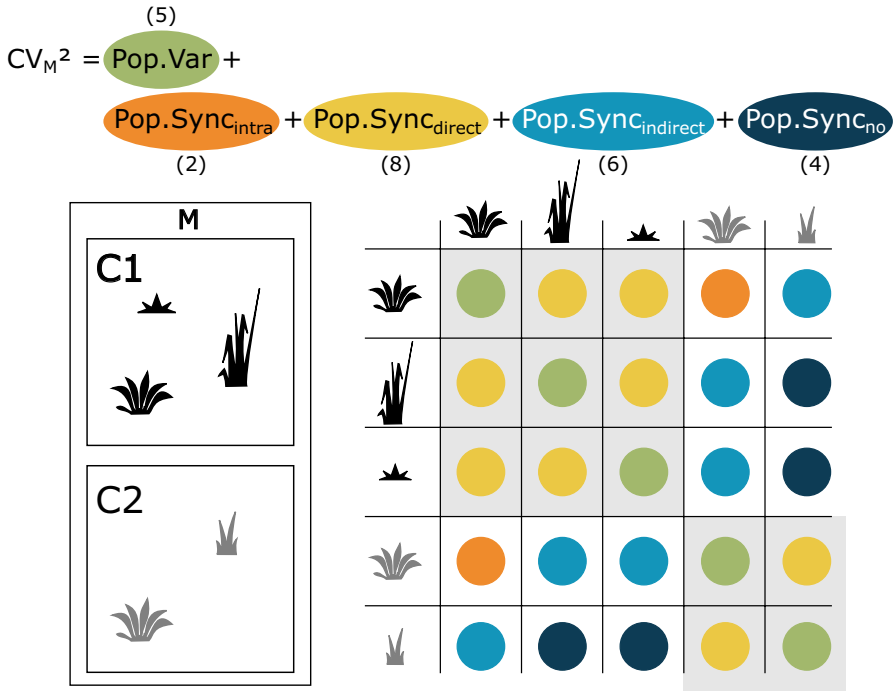


FIGURE 1 Graphical representation of the partitioning of the metacommunity squared coefficient of variation (CV_M^2) into *Pop.var* and the four *Pop.syncs* (see Equations 3 and 4). In this example, the metacommunity (M) contains two communities (C1 and C2) with a different number of species (represented by different icons): C1 has three species, C2 has two species, and they share one species. The grid represents the variance–covariance matrix between the five populations. The colours indicate the terms included in each CV_M^2 component: In the diagonal, the variance terms (green circles) are used to compute *Pop.var*, while covariance terms are sorted according to the population identities and used to compute *Pop.syncs* within species (orange circles, $Pop.sync_{intra}$), among species within a community (yellow circles, $Pop.sync_{direct}$), among species among communities with indirect interaction (blue circles, $Pop.sync_{indirect}$) and no interaction (dark blue circles, $Pop.sync_{no}$). The grey areas represent the within-community components, while white areas identify the terms used to scale up to the metacommunity level. In the equation, the number of terms in each CV_M^2 component is given in brackets

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2 | MATERIALS AND METHODS

2.1 | General framework

We define a metacommunity (M) as a group of *m* spatially defined communities. For analytical reasons, we consider the metacommunity as a set of the *n* local populations (independently of the species identity) monitored in the *m* communities. Here a population corresponds to the individuals of a species growing in a community. Figure 1 shows a metacommunity composed of $m = 2$ communities or $n = 5$ local populations. The annual metacommunity biomass is the sum of the *n* local population biomasses produced each year. The temporal variability of the metacommunity productivity is calculated as the squared coefficient of variation (CV_M^2) of the annual metacommunity biomass over several years:

$$CV_M^2 = \frac{var_M}{\mu_M^2} \quad (1)$$

where var_M and μ_M are the variance and mean of the annual metacommunity biomass. We propose a framework that decomposes CV_M^2 into local populations effects. The total metacommunity variance can be decomposed into the independent fluctuations of individual populations and their covariances as follow:

$$CV_M^2 = \frac{1}{\mu_M^2}\left(\sum_{i=1}^{n} var_i + \sum_{i=1}^{n}\sum_{j\neq i}^{n} cov_{i,j}\right) \quad (2)$$

where var_i and $cov_{i,j}$ are the biomass variance and covariance of the *n* local populations of the metacommunity (i *and* $j \in [1: n]$). The latter can be rearranged as the sum of two terms with contrasting ecological significances:

$$CV_M^2 = \sum_{i=1}^{n}(p_i\, CV_i)^2 + \frac{1}{\mu_M^2}\sum_{i=1}^{n}\sum_{j\neq i}^{n} cov_{i,j} = \text{Pop. var} + \text{Pop. sync} \quad (3)$$

where p_i and CV_i are, respectively, the relative abundance ($p_i = \mu_i/\mu_M$) and the coefficient of variation of population *i* and $cov_{i,j}$ is the temporal covariance of the local populations *i* and *j*. The first term (*Pop.var*) corresponds to the effect of population variability. It is the expected value of CV_M^2 under independent fluctuation between populations. *Pop.var* can be written as the square of individual population *CV* weighted by their abundance (the variability of abundant populations mostly determines *Pop.var*). The second term (*Pop.sync*) is the effect of population synchrony and can be positive (prevalent synchrony), null (independent fluctuations) or negative (prevalent anti-synchrony). It follows a key point presented in the introduction about measuring positive, negative or independent synchrony, rather than a deviation from perfect synchrony (Box 1). Since both *Pop.var* and *Pop.sync* terms are standardized by the metacommunity biomass, rare species contribute little to CV_M^2.

Pop.sync can be further decomposed into four different additive terms (see Figure 1 for a graphical representation) according to the identities of the two populations (*i* and *j*) involved in the covariance terms:

$$\text{Pop. sync} = \frac{1}{\mu_M^2}\sum_{i=1}^{n}\sum_{j\neq i}^{n} cov_{i,j} = \frac{1}{\mu_M^2}\sum_{i=1}^{n}\sum_{j\neq i}^{n}\alpha_{i,j}.cov_{i,j} + \frac{1}{\mu_M^2}\sum_{i=1}^{n}\sum_{j\neq i}^{n}\beta_{i,j}.cov_{i,j} + \frac{1}{\mu_M^2}\sum_{i=1}^{n}\sum_{j\neq i}^{n}\gamma_{i,j}.cov_{i,j} + \frac{1}{\mu_M^2}\sum_{i=1}^{n}\sum_{j\neq i}^{n}\delta_{i,j}.cov_{i,j}$$

$$= \text{Pop. sync}_{\text{intra}} + \text{Pop. sync}_{\text{direct}} + \text{Pop. sync}_{\text{inderect}} + \text{Pop. sync}_{\text{no}} \quad (4)$$

where

$$\alpha_{i,j} = \begin{cases} 1 \text{ if } S_i = S_j \text{ and } C_i \neq C_j \\ 0 \text{ otherwise} \end{cases}$$

$$\beta_{i,j} = \begin{cases} 1 \text{ if } S_i \neq S_j \text{ and } C_i = C_j \\ 0 \text{ otherwise} \end{cases}$$

$$\gamma_{i,j} = \begin{cases} 1 \text{ if } S_i \neq S_j \text{ and } C_i \neq C_j \text{ and } (S_i \in C_j \text{ or } S_j \in C_i) \\ 0 \text{ otherwise} \end{cases}$$

$$\delta_{i,j} = \begin{cases} 1 \text{ if } S_i \neq S_j \text{ and } C_i \neq C_j \text{ and } (S_i \notin C_j \text{ and } S_j \notin C_i) \\ 0 \text{ otherwise} \end{cases}$$

S_i and S_j are the species identity of population *i* and *j*, respectively, and C_i, C_j, the communities where population *i*, *j*, occurred. The within species synchrony ($Pop.sync_{intra}$) includes the covariance terms involving populations of the same species ($S_i = S_j$) in different communities ($C_i \neq C_j$). The direct interaction synchrony ($Pop.sync_{direct}$) includes covariance terms involving populations of different species ($S_i \neq S_j$) within a community ($C_i = C_j$). The indirect interaction synchrony ($Pop.sync_{indirect}$) includes covariance terms involving populations of different species ($S_i \neq S_j$) in different communities ($C_i \neq C_j$) but where both species occur together in at least one community (S_i *in* C_j *and/or* S_j *in* C_i). Finally, the no interaction synchrony ($Pop.sync_{no}$) includes covariance terms involving populations of different species ($S_i \neq S_j$) in different communities ($C_i \neq C_j$) but where both species occur in their community only (S_i *not in* C_j *and* S_j *not in* C_i). The number of covariance terms in each synchrony term depends on the experimental design, the metacommunity species richness and the species turnover between communities (Figure 1).

$Pop.sync_{intra}$ describes the overall synchrony between populations of the same species growing in different communities and is expected to be positive under the hypothesis of a common response to environmental fluctuations (Table 1). $Pop.sync_{direct}$ describes the overall synchrony between populations that are interacting locally and is expected to be negative in the case of high competition between dominant species (Table 1). $Pop.sync_{indirect}$ evaluates the intensity of indirect competition across communities, especially compared to $Pop.sync_{direct}$, and $Pop.sync_{no}$ the coordination of species responses to environmental conditions (Table 1).

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R code to run the partitioning of the metacommunity squared coefficient of variation is available at the following URL: https://github.com/jsegrestin/MetacomStab.

2.2 | Data collection

The study site is located 10 km southeast of České Budějovice, Czech Republic (48°570N, 14°360E, altitude 510m). The vegetation is an oligotrophic wet species-rich meadow dominated by grasses, especially *Molinia caerulea* (L.) Moench. It is representative of the extensively mown meadow of the surrounding region. In 1994, 12 permanent plots of 2×2 m were established. Fertilization and removal of *M. caerulea* were applied as treatments in a factorial design, with three replicates per treatment combinations (FR: fertilization and removal, FU: fertilization and no removal, UR: no fertilization and removal, UU: no treatment). In the fertilization treatments, 65g/m^2 of a commercial NPK fertilizer (12% N, 19% P, 19% K) was added annually. Since 2003, the fertilizer was changed to Cererit (8% N, 13% P, 11% K), and was added at a lower dose of 50g/m^2 since 2012, to prevent nutrient accumulation in the soil, particularly phosphorus. In the removal treatments, *M. caerulea* was manually weeded in 1995 and new individuals growing afterward were removed annually. More details on the experimental design are given in Lepš et al. (2018) and references therein.

After 5years, the community composition stabilized in all treatments. During the 1999–2014 period, species biomass in the central 0.25m^2 of each plot was recorded each year in the second half of June. Biomass was clipped and was sorted into species, oven-dried and weighed. More technical details of biomass sorting of these extremely species-rich communities are described in Appendix 1 of Lepš et al. (2018). The clipping of the central part of the plots was applied simultaneously to the annual mowing of the site (including the rest of the plots). We therefore monitored the exact same plot each year. *Molinia caerulea* was often found in small quantity in the removal treatment and were not remove from the data. The final dataset describes the temporal variation of the productivity of 600 plant populations of 70 species for 16years (average of 50 populations per community).

2.3 | Data analyses

Each year, the productivity of plots, treatments and the total metacommunity is calculated as the sum of the corresponding population biomasses. To study the effect of the spatial scale on CV^2, we applied our general framework to different subsets of the metacommunity. We computed the biomass CV^2 of individual populations (n = 600), communities (plot, n = 12), treatments (n = 4) and metacommunity (n = 1) using Equation 1. At the community, treatment and metacommunity levels, we computed *Pop.var* and *Pop.sync* values using Equation 3 with the corresponding populations. In this analysis, *Pop.var* values show the mathematical decrease in CV^2 with the spatial scale. Indeed, *Pop.var* of the metacommunity is mathematically lower than the *Pop.var* of subsamples of that metacommunity on average, capturing the spatial 'portfolio effect' proposed by Doak et al. (1998). *Pop.sync* values show the overall synchrony effect at the different spatial scales.

The effect of the spatial scale was also tested at the intraspecific level. We selected four abundant species of the metacommunity (two grasses and two forbs), namely *M. caerulea*, *Festuca rubra* L., *Potentilla erecta* (L.) Räusch. and *Cirsium palustre* (L.) Scop. These four species represent 17% to 48% of the community biomass depending on the plot. They are perennial species found in the 12 plots, apart from *M. caerulea* that was weeded in the removal treatments. Both grasses are polycarpic tussocks, *P. erecta* is a polycarpic herb and *C. palustre* is a perennial monocarpic rosette. We calculated the CV^2 of populations (n = 12 or 6), treatments (n = 4 or 2) and the metapopulation (n = 1), for each species separately. Similarly, *Pop.var* and *Pop.sync* were calculated at the treatment and the metapopulation levels.

Finally, we decomposed the metacommunity CV_M^2 into additive subgroups of *Pop.vars* and *Pop.syncs*. We calculated four *Pop.vars* corresponding to the four treatments (using the variance terms of the corresponding populations). *Pop.sync* was decomposed into $Pop.sync_{intra}$, $Pop.sync_{direct}$, $Pop.sync_{indirect}$ and $Pop.sync_{no}$ (Equation 4, Figure 1) for each treatment and $Pop.sync_{intra}$, $Pop.sync_{indirect}$ and $Pop.sync_{no}$ between treatments (using the covariance terms of the corresponding populations). For *Pop.syncs*, we used a randomization procedure to test whether the observed values were significantly different from independent fluctuations between populations. The procedure consists of 10,000 randomizations of the dataset, where the temporal sequence of biomass values of each population is randomized. The randomization does not affect the CV^2 of individual population but simulates independent fluctuations between them. Then, we computed values of *Pop.syncs* on the 10,000 randomized datasets. We used the 0.025 and 0.975 quantiles of the resulting values as the range of independent fluctuations between populations (*Pop.sync* not different from 0). Moreover, *Pop.syncs* values were compared to their maximum value under perfect synchrony. This value is calculated by replacing all covariance terms by the product of standard deviations. Due to the strong differences in the number of terms summed in each effect, the range of independent fluctuation and the maximum value under perfect synchrony can be very different between synchrony components. Note that the minimum value for each subgroup of *Pop.sync* cannot be calculated. Indeed, although the total *Pop.sync* cannot be lower than—*Pop.var* (when $CV_M^2 = 0$), this value can be achieved by multiple combinations of values of *Pop.sync* subgroups.

3 | RESULTS

3.1 | Effects of the spatial scale

All local populations but one (n = 599) had higher CV^2 than the community in which they occur (Figure 2a): one population of *M. caerulea* in the fertilized and removal treatment had a CV^2 of 0.046 while the total community had a CV^2 of 0.053. This is an apparent artefact of the treatments because this population is represented by very few individuals able to resprout despite the frequent weeding. Moreover, the CV^2 (mean ± *SD*) decreased with the spatial scale (Figure 2a):

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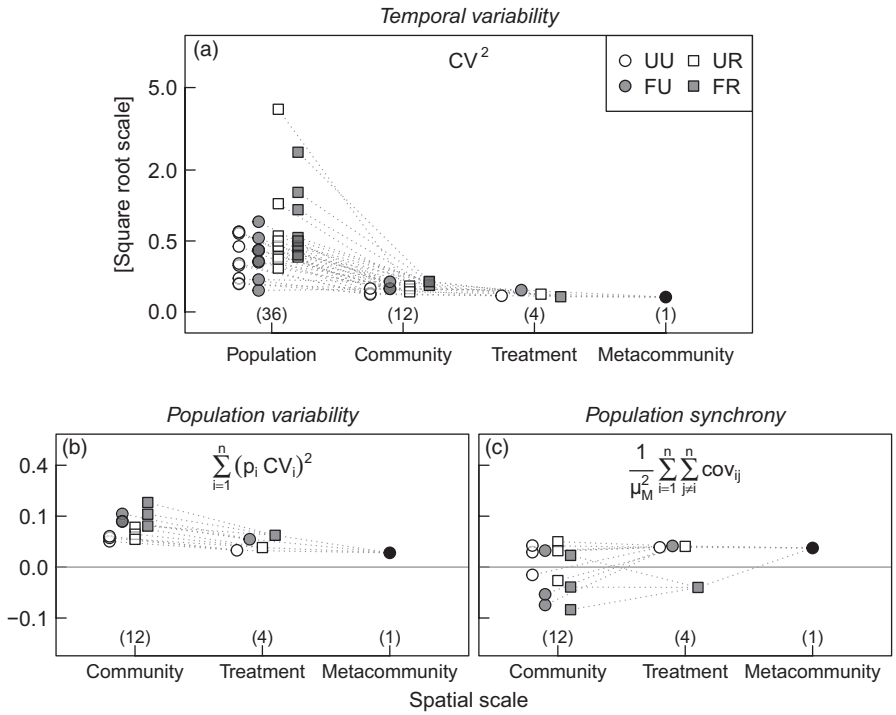


FIGURE 2 (a) Squared coefficient of variation (CV^2) of the annual biomass production over the 1999–2014 period at different spatial scales. For readability, only the populations of the three most abundant species in each community are represented. At the scales of communities, treatments and metacommunity, each CV^2 is decomposed into two additive terms: (b) *Pop.var* and (c) *Pop.sync* (see Equation 3 for details). The treatments are UU: Unfertilized and no species removal, FU: Fertilized and no species removal, UR: Unfertilized and *M. caerulea* removal and FR: Fertilized and *M. caerulea* removal. The number of points in each category is in brackets

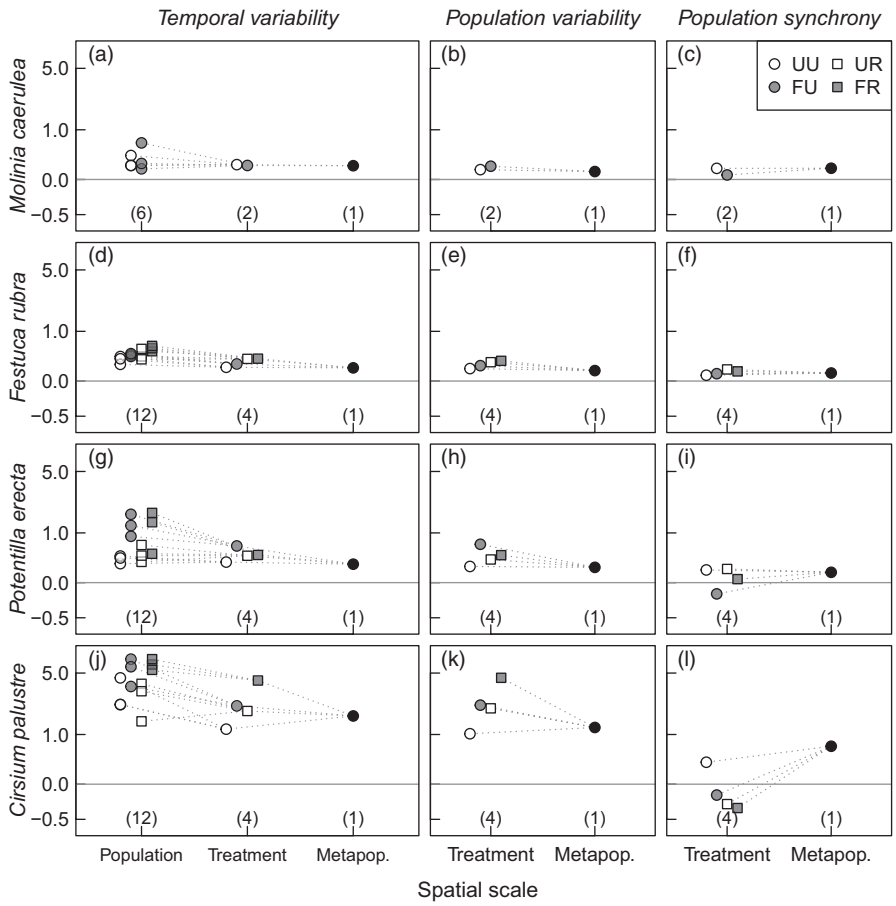


FIGURE 3 (a, d, g, j) Squared coefficient of variation (CV^2) of the annual biomass production of four selected species over the 1999–2014 period at different spatial scales. At the scales of treatments and metapopulation, each CV^2 is decomposed into two additive terms: (b, e, h, k) *Pop.var* and (c, f, i, l) *Pop.sync* (see Equation 3 for details). The treatments are UU: Unfertilized and no species removal, FU: Fertilized and no species removal, UR: Unfertilized and *M. caerulea* removal and FR: Fertilized and *M. caerulea* removal. The number of points in each category is in brackets

0.061 ± 0.022 at the community level (n = 12), 0.032 ± 0.011 at the treatment level (n = 4) and 0.022 at the metacommunity level (n = 1). It results from the mathematical decrease in *Pop.var* with an increasing number of populations (Figure 2b): *Pop.var* = 0.069 ± 0.041 at the community level, 0.024 ± 0.013 at the treatment level and 0.008 at the metacommunity level. The *Pop.sync*, on the other hand, tends

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to counterbalance this mathematical trend. We found negative and positive values at a small spatial scale and the highest positive value found at the metacommunity level: *Pop.sync* = −0.008 ± 0.029 at the community level, 0.008 ± 0.016 at the treatment level and 0.014 at the metacommunity level (Figure 2c). When compared to randomizations, one community in FU was significantly anti-synchronous, while both unfertilized treatments and the metacommunity were significantly synchronous (Figure 2c, Table S1). All other communities and treatments had an overall synchrony that was not significantly different from independent fluctuations (Table S1).

Similar trends were found at the intraspecific level in four dominant species (Figure 3). When comparing species, CV^2 increased in the following order: *M. caerulea* < *F. rubra* < *P. erecta* < *C. palustre* at all spatial scales (Figure 3a,d,g,j). The same conclusion applies for *Pop.var* (Figure 3b,e,h,k). However, *Pop.sync* showed reversed patterns at the treatment level compared to the metapopulation level (Figure 3c,f,i,l).

3.2 | Partitioning of the metacommunity CV_M^2

Pop.var values in fertilized treatments were higher than those in unfertilized treatments (Figure 4). Accordingly, the range of independent fluctuations between populations, the maximum values under perfect synchrony and the observed *Pop.sync* values had higher amplitude in fertilized treatments than those in unfertilized treatments (Figure 4). Although close to boundary values, *Pop.sync* components were mainly included in the range of independent fluctuations between populations. Significant positive synchrony was found within species in each treatment separately and between treatments, and between species between treatments ($Pop.sync_{intra}$ in Figure 4). Significant but weak anti-synchrony was found between non-interacting species in the unfertilized and unremoval treatment ($Pop.sync_{indirect}$ in Figure 4). Strongly negative values of $Pop.sync_{direct}$ in Figure 4). Although included in the range of independent fluctuations, these values decreased significantly the overall CV_M^2. These results were confirmed in a cross-validation procedure (using 100 sub-datasets corresponding to random selections of 10 among the 16 years of survey) to make sure that the synchrony patterns were not strongly affected by a reduced number of extreme years (Figure S1).

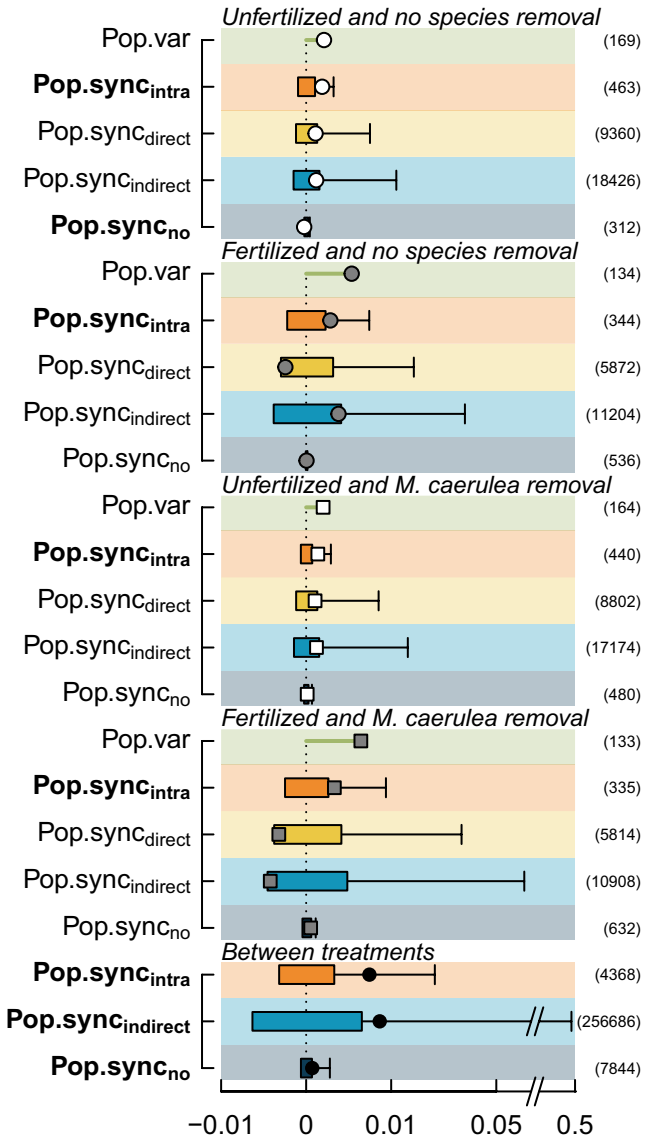


FIGURE 4 Partitioning of the squared coefficient of variation of the metacommunity productivity (CV_M^2) over the 1999–2014 period, into *Pop.var* and *Pop.sync* components (see Equations 3 and 4 for details). The symbols represent the observed values in each treatment and between treatments. The total sum of the observed values is $CV_M^2 = 0.022$. For the *Pop.sync* components, coloured boxes show the range of expected values under independent fluctuation of populations (95% intervals calculated from randomizations) and the whiskers represent the maximum value under perfect synchrony of populations. *Pop.syncs* out of the boxes (significantly different from zero) are in bold. Positive or negative values represent significant synchrony or anti-synchrony between populations respectively. Data are represented on a square root axis that is broken for readability. The number of terms summed in each component is in bracket. Note that the distance between the observed values and the tip of the whiskers correspond to the indices proposed by Hammond et al. (2020) (see Box 1)

4 | DISCUSSION

Our framework aims at disentangling the relative effects of several ecological processes on productivity stability, including spatial stabilization, species responses to environmental fluctuations, species demography and species interactions. Since the same treatment was applied each year at the community level, plant populations experienced low interannual variations in nutrient availability or disturbance during the experiment. Similarly, we did not observe strong annual fluctuations in population attacks by plant herbivores or pathogens. Therefore, these factors were not considered. In previous analyses (Lepš, 2004; Lepš et al., 2018), we demonstrated the

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effects of the treatments on the stability of individual (local) populations and communities. We found that fertilization decreases the synchrony between species but strongly increases their variability. Consequently, the higher asynchrony did not compensate for the increase in species variability, and fertilized communities were less stable than unfertilized ones. Here, we scale up to the metacommunity and focus on the determinants of its stability using the treatments as a simulation of spatial environmental heterogeneity. In the following parts, we investigate (1) the patterns and drivers of the spatial stabilization between and within species and (2) the ecological interpretations of synchrony indices across organizational and spatial scales.

4.1 | Insights on the spatial stabilization

It has been proposed that the temporal stability of ecosystem functions should relate positively with the spatial scale (Delsol et al., 2018; Peterson et al., 1998). Our study, associated with other empirical examples (Wang et al., 2017; Wang & Loreau, 2014), supports this hypothesis since CV^2 decreased with spatial scale. This pattern has been related to the species–area relationship (Wang et al., 2017; Zhang et al., 2018), which predicts higher species diversity at a larger spatial scale (Rosenzweig, 1995). In turn, a higher diversity is likely to result in higher stability through two major processes: (1) the mathematical stabilization due to a higher number of fluctuating entities, introduced by Doak et al. (1998) as the portfolio effect and (2) the lower synchrony between populations due to the higher probability of including species with dissimilar functioning, resulting in different responses to environmental fluctuations (the 'insurance hypothesis' Naeem & Li, 1997; Rocha et al., 2011; Yachi & Loreau, 1999; Table 1). Although the 'insurance hypothesis' can decrease the synchrony between species, it is unlikely to result in anti-synchronous patterns (Appendix S2). In our study, the spatial variation of *Pop.var* captured the portfolio effect and was the major driver of the decrease in CV^2 with spatial scale. However, and contrary to the predictions, population synchrony (*Pop.sync*) tended to increase with spatial scale, counterbalancing spatial stabilization. We observed signals of anti-synchrony in local communities, while populations were mostly synchronized at the metacommunity level. Therefore, we show no evidence for a reduction of synchrony at a larger spatial scale. One explanation can be the small increase in species richness with space in the studied meadow. Although the metacommunity had the highest species richness (70 species), the system had low β diversity (50 species at the community level and 62.5 species at the treatment level on average). Moreover, the species not found in all plots were rare in the community where they occurred and therefore had little effect on the metacommunity stability. Whether the same conclusions apply in other systems including larger species turnover between plots remains to be tested. With higher species turnover, the metacommunity would include more pairs of non-interacting populations, which are expected to be close to independent fluctuations, especially when they belong to species with contrasting responses to environmental cues.

Increasing spatial scale can reduce synchrony between populations by integrating environmental heterogeneity (Hammond et al., 2020; Wang et al., 2019; Wilcox et al., 2017). Indeed, we can expect independent or anti-synchronous fluctuation between populations from plots with different resource availability or disturbance regime (Table 1). Heisler and Knapp (2008) reported that the temporal correlation between plot productivity (reported as coherence) of a mesic grassland depended on soil depth and fire frequency. In our study, the environmental heterogeneity simulated by the treatments did not result in lower population synchrony across communities. The overall synchrony between populations in homogenous space (within treatments) was close or even lower than the synchrony between populations in heterogeneous space (between treatments). One explanation can be the small heterogeneity in the disturbance regime or soil characteristics in our experiment.

To our knowledge, this is the first study investigating explicitly the effect of environmental heterogeneity on the synchrony of population fluctuations. We used experimentally manipulated communities to simulate the functioning of a grassland metacommunity comprising different levels of nutrient availability. It enabled a straightforward analysis of the heterogeneity effect by comparing synchrony within and between treatments. Nevertheless, future studies need to compare our results with metacommunities under natural conditions, including larger environmental heterogeneity and at a larger spatial scale (e.g. regional level with multiple habitats). Although not appropriate to our experimental design, that prevent the spatial clustering of plots under the same treatment, further development of our framework for natural metacommunities could incorporate a more spatially explicit analysis, exploring how the distance between communities affects the population synchrony, as introduced by Wang et al. (2017).

4.2 | What drives metapopulation stability?

When considering all plots and all species, and in agreement with our hypotheses, we found significantly positive $Pop.sync_{intra}$. It probably relates to similar responses of plant populations within species to environmental fluctuations due to their similar functioning. If the intraspecific synchrony is close to perfect synchrony, the metapopulation variability is close to the weighted-mean population variability ($CV^2 \approx \left(\sum_i p_i CV_i\right)^2$). Therefore, individual population variability, and not population synchrony, is the major driver of the overall metapopulation stability (Polley et al., 2007; Sasaki & Lauenroth, 2011). Consequently, the stability of metapopulations should mainly relate to plant traits that influence individual population variability.

Interestingly, *M. caerulea*, the most abundant species in the studied meadow, was also the most stable species among the 70 collected species. This is probably related to its overwinter carbohydrates reserves stored in basal internodes (Bartoš et al., 2011; Janeček et al., 2004) allowing some decoupling between climate

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conditions and growth in spring (storage effect) and its high rooting depth (Kutschera & Lichtenegger, 1992) allowing the reach of underground water table even during the driest years. Conversely, *C. palustre*, which also belongs to the dominant species, had one of the highest CV^2, especially at the population scale. *C. palustre* is a large monocarpic perennial with a short life span. Each population is therefore composed of a few individuals for which annual growth depends mostly on their life stage. The result is a high temporal variability of population biomass but low synchrony between populations due to demographic stochasticity (de Mazancourt et al., 2013 and see Table 1). With increasing spatial scale, the effect of the demographic stochasticity is buffered by the higher number of individuals, resulting in a lower effect of individual population variability but higher spatial synchrony. Although not tested here, other plant traits can influence population stability. For example, Májeková et al. (2014) found a positive relationship between leaf dry matter content (negative predictor of species growth rate) and population stability.

Surprisingly, the intraspecific synchrony was higher when considering populations from different treatments than within treatments, challenging the idea that spatial heterogeneity could disrupt synchrony and stabilize metapopulations (Table 1). Whether the same conclusions apply in other systems, especially with higher environmental heterogeneity, remains to be tested. For example, differences in soil characteristics between plots could create a decoupling between precipitations and water availability for plants resulting in lower synchrony at the intraspecific level.

4.3 | What are the ecological interpretations of positive, null and negative synchrony?

Here, we consider that an overall positive, negative or independent synchrony between populations reflects the main ecological processes driving the compensation between population fluctuations. In our framework, synchrony values are mostly driven by the behaviour of the most dominant species and relate to their responses to environmental fluctuations, demography or interactions (Table 1). Although an overall independent and positive synchrony can be found at many organizational and spatial scales, anti-synchrony is theoretically restricted to competitive interactions (Lepš et al., 2019; Tilman et al., 1998), cases of strong spatial heterogeneity or attacks by specialist enemies (Kranz, 1990; Table 1). In our system, the anti-synchrony related to interspecific competition likely results from the storage effect (Chesson, 2000) where (1) species have differential responses to the environment, (2) the competition outcome depends on the annual environmental conditions and (3) there is a mechanism that buffers population numbers when the environment is unfavourable for a species (see Chesson, 2000; Gonzalez & Loreau, 2009). For example, anti-synchrony can be expected between species with different drought tolerance if the annual water availability modifies the outcome of their competitive interactions (e.g. a dry year releases the competitive pressure on drought-tolerant species), and they can all persist even when outcompeted through seed bank, dormancy, reduced density or immigration. The fact that anti-synchrony was found between interacting populations strengthens the idea that competition leads to strong compensatory dynamics, especially in fertilized communities. In agreement with theoretical and empirical results (Ives et al., 1999; Tilman, 1996), these compensatory dynamics were not associated with higher community stability due to the higher destabilizing effect of competition on individual populations (Lepš, 2004; Lepš et al., 2018). On the other hand, cross-community synchrony was significantly positive (within and between species) when considering populations from different treatments. The three synchrony terms between treatments increased the CV_M^2 by 2.4 times in comparison with the expected value under independent fluctuations and, in that sense, can be considered as major destabilizing factors. Overall, our analysis suggests that positive synchrony among plant populations is prevalent at the metacommunity level, probably due to common responses to climate conditions, but is mitigated at the local scale by species interactions. These observations are consistent with the analysis of Houlahan et al. (2007) showing that anti-synchrony is rare in a large set of communities and is found in the smallest plots. Studies at the metacommunity level are still scarce and our conclusions need to be compared with other systems including larger spatial scale and environmental heterogeneity.

Based on our analyses, we conclude that cross-community synchrony tends to destabilize the metacommunity, while anti-synchronous fluctuations are found locally. Our conclusions contrast with those of Hammond et al. (2020), which found that cross-community asynchrony is the most stabilizing factor of the metacommunity. These discrepancies result from a major conceptual difference between frameworks. More precisely, our indices focus on synchrony patterns (related to ecological hypotheses in Table 1), while the whole compensatory dynamics (sensu Gonzalez & Loreau, 2009) evaluated by the indices from Hammond et al. (2020) also include the effect of the population number and variability (Appendix S1). Based on our data, $Pop.sync_{indirect}$ between treatments is the most synchronous factor using our framework (highest sum of covariances) but is also, by far, the most asynchronous factor using the framework from Hammond et al. (2020) (strongest deviation from perfect synchrony) due to the very high number of population pairs considered (Figure 4, Figure S2). Moreover, the indices presented in Hammond et al. (2020) could prevent, in some cases, distinguishing anti-synchrony from positive synchrony due to the large difference in population number used in each category. In our study in FU, $Pop.sync_{direct}$ is anti-synchronous and $Pop.sync_{indirect}$ is synchronous, while the corresponding asynchrony indices from Hammond et al. (2020) show the reverse pattern ($\delta < \beta_{CCi}$, Figure S2) although calculated with the same populations. It confirms that the indices proposed by Hammond et al. (2020) are not suitable to detect signals of ecological processes, since they are more affected by the intrinsic characteristics of the metacommunity (number of communities and their species richness) than signals of positive or negative synchrony. These counterintuitive results reveal that the current debate over the most accurate benchmark for synchrony

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has profound effects on our understanding of the processes underlying the stability of metacommunities. Although φ related indices are reliable to estimate the overall stabilization compared to perfect synchrony, our framework appears as a powerful tool to disentangle various ecological processes affecting simultaneously metacommunity stability.

5 | CONCLUSIONS

We demonstrated that the spatial stabilization of a temperate meadow was mainly driven by the portfolio effect, and we found no signal of a lower synchrony between populations at a larger spatial scale. The species identity had a major effect on population and metapopulation stability. Our analysis suggests that positive synchrony among plant populations prevails at the metacommunity level but is mitigated at the local scale by species interactions. Whether the same conclusions apply in systems with larger environmental heterogeneity and β diversity is a promising perspective for further investigation, especially under natural conditions. We also demonstrated that the choice of synchrony indices should be associated with cautious interpretations, revealing the urgent need for analytical guidelines to study ecosystem stability in the future, especially at the metacommunity level.

ACKNOWLEDGEMENTS

This research was supported by GAČR 20-02901S. We thank the many students and volunteers helping with biomass sampling and sorting during all the years and Pavel Fibich for helping with formula simplifications in Mathematica.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

PEER REVIEW

The peer review history for this article is available at https://publons.com/publon/10.1111/1365-2745.13898.

AUTHORS' CONTRIBUTIONS

J.L. conceived the idea, designed the methodology, conducted the experiment and collected the data; J.S. developed the analytical framework, run the analyses and led the writing of the manuscript. Both authors contributed critically to the drafts and gave final approval for publication.

DATA ACCESSIBILITY STATEMENT

Data are deposited in the Dryad Digital Repository https://doi.org/10.5061/dryad.8gtht76r9 (Segrestin & Lepš, 2022). R code to run the partitioning of the metacommunity squared coefficient of variation is available at the following URL: https://github.com/jsegrestin/MetacomStab and have been archived via Zenodo (Segrestin, 2022).

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How to cite this article: Segrestin, J. & Lepš, J. (2022). Towards a better ecological understanding of metacommunity stability: A multiscale framework to disentangle population variability and synchrony effects. *Journal of Ecology*, *110*, 1632–1645. https://doi.org/10.1111/1365-2745.13898

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